

HELIX Toolbox — Spall Detection & Qualification

This note describes the spall-detection and did-not-spall (DNS) qualification logic as implemented in the HELIX Toolbox, and explains every tuning parameter with a graphic example on a real 128 GS/s Ti PDV trace.

What the algorithm does

ALPSS → free-surface velocity (Savitzky–Golay smoothed) + uncertainty

1. Uncertainty filter drop points with rel. uncertainty ≥ 1.0
2. Align $t = 0$ to ALPSS shock arrival (signal start)
3. Window keep spall_start ... spall_end ns
4. Gaussian smooth one pass, spall_smoothing_sigma_ns
5. Plateau points $\geq$ plateau_threshold $\times$ peak → P1, P2, mean
6. Pullback valley P3 first prominent minimum after the plateau ...
7. Recompression P4 ... that is FOLLOWED BY a prominent peak (local)
8. Gates 4 recompression checks as a final refinement
→ first valley whose recompression passes = VALID SPALL, else DNS

The spall signature

A spall reflects a wave off the newly-formed internal fracture surface, so the measured velocity shows a two-sided fingerprint: a pullback dip (P3) followed by a recompression rebound (P4). Requiring BOTH — a prominent valley AND a prominent following peak — is the core of the qualification. A trace that simply shocks and releases (no rebound) is DNS.

Physics (VALID spall traces only)

Spall strength $\sigma = \frac{1}{2} \cdot \rho \cdot c_b \cdot \Delta u_{\text{pullback}}$ ($\Delta u = \text{plateau} - P3$)
Strain rate $\dot{\epsilon} = |\text{release slope}| / (2 \cdot c_b)$
Shock stress $\sigma = \rho \cdot U_s \cdot u_p$, $U_s = c + S \cdot u_p$, $u_p = \text{plateau}/2$

DNS update: P3 velocity must be > 0 m/s

Any trace with a pullback minimum P3 at zero or below zero velocity is DNS.

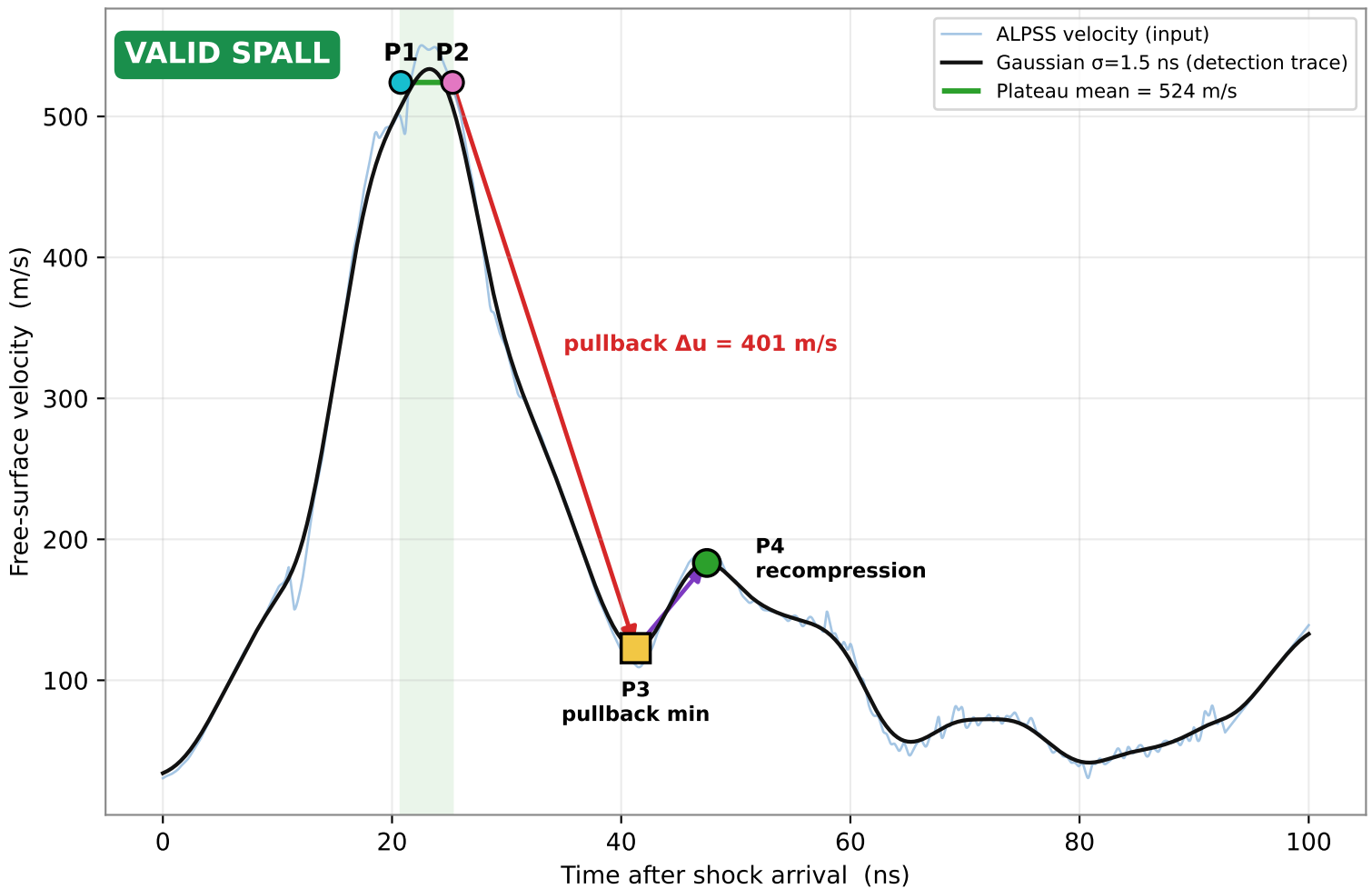
This physical criterion is checked before the P4 recompression gates.

No spall strength is reported for these traces; the diagnostic fit is retained.

The following pages: (2) the algorithm on a real trace, (3) a parameter summary, then one graphic per tuning parameter.

The algorithm on a real trace (Ti)

JHAMAL00016-004_2026-04-23_21-52-09_shot20_ch1 — $\rho=4507 \text{ kg/m}^3$, $c=4726 \text{ m/s}$, $S=1.049$



What the code computed here

Plateau	P1-P2	mean = 524 m/s	→ shock stress
Pullback	P3	123 m/s at 41.3 ns	$\Delta u = 401 \text{ m/s}$ → spall strength
Recomp.	P4	183 m/s at 47.5 ns	rebound = 60 m/s

All four recompression gates pass: $P4 > P3$ ($183 > 123$); $P4 \geq P3 \times 1.015$ ($183 \geq 125$);

$t(P4) - t(P3) \geq 2.5 \text{ ns}$ (6.2 ns); rebound $\geq 0.015 \times \text{pullback}$ ($60 \geq 6.0$).

→ $\sigma_{\text{spall}} = 4.275 \text{ GPa}$, $\dot{\epsilon} = 2.66\text{e}+06 \text{ s}^{-1}$, $\sigma_{\text{shock}} = 5.907 \text{ GPa}$

Tuning parameters — summary

All live in the 'SPALL DETECTION FILTERS' block of the config

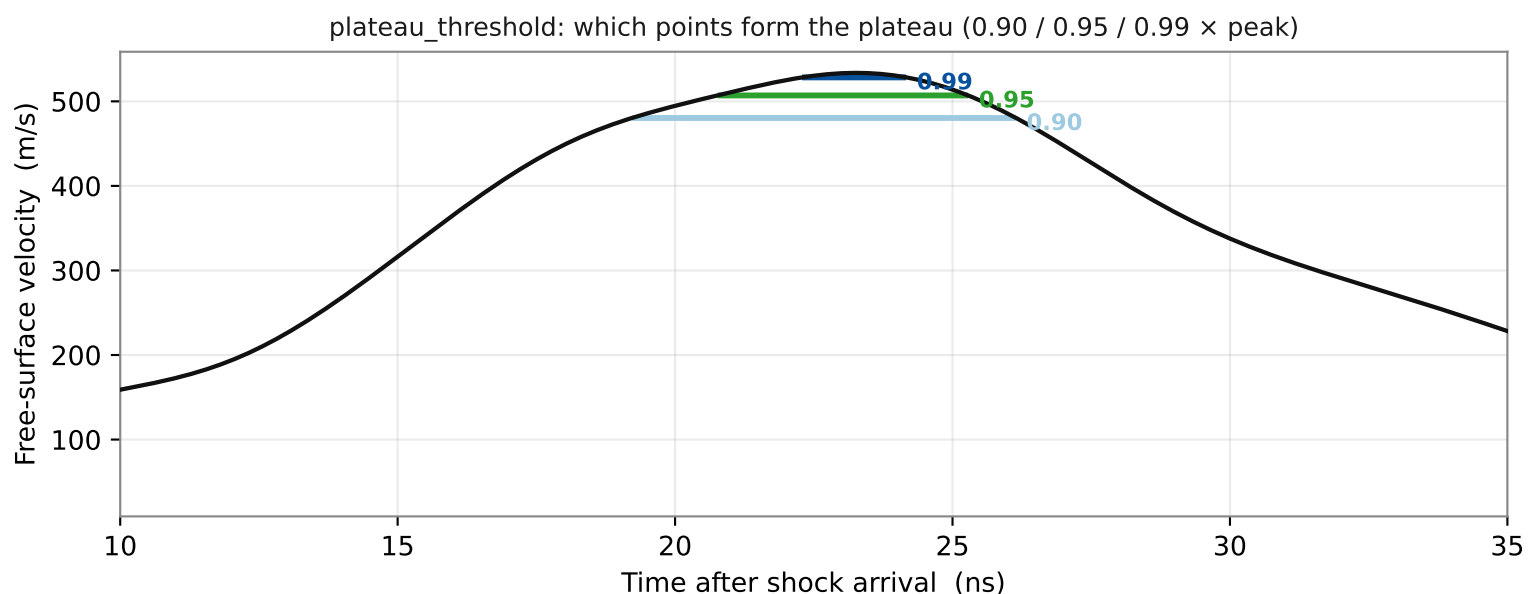
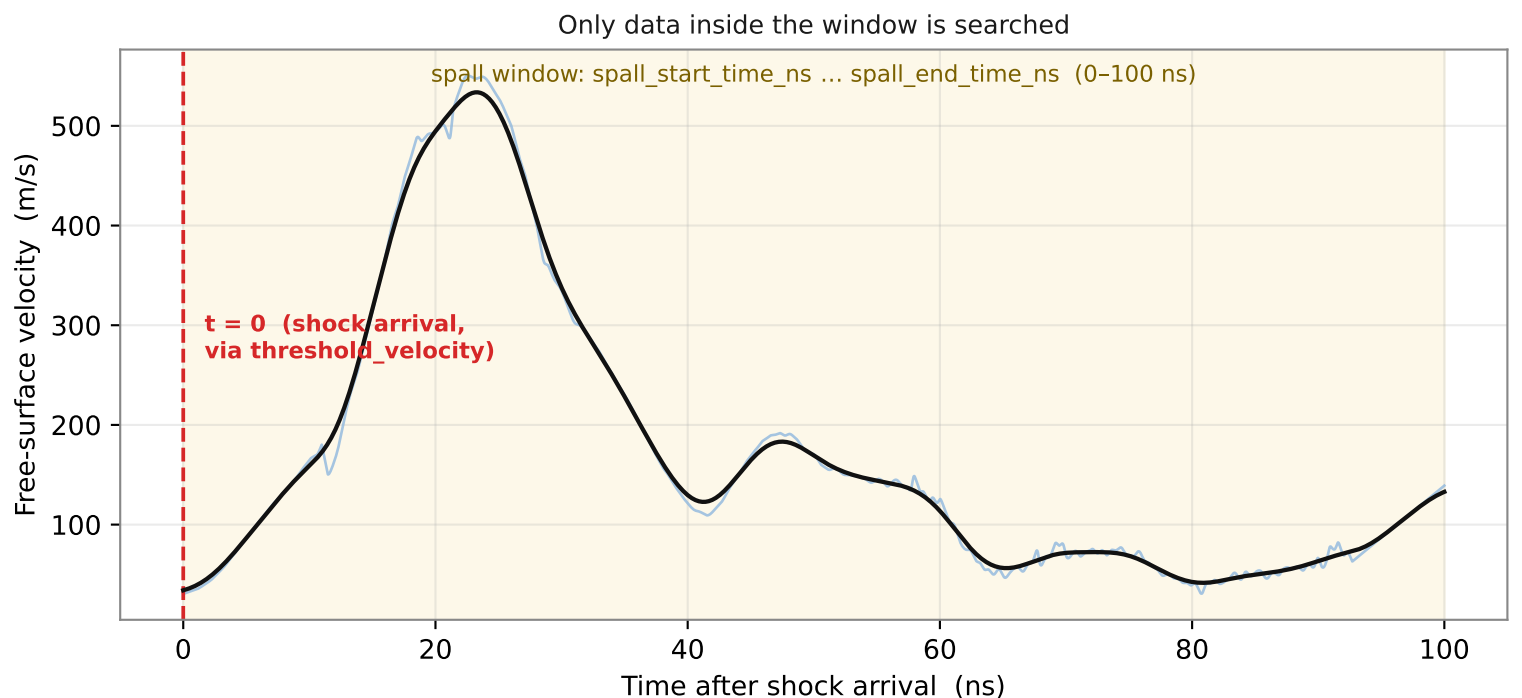
Parameter	Default	Controls
plateau_threshold <i>↑ narrower plateau (near peak); ↓ wider, includes rise/fall</i>	0.95	fraction of peak that counts as plateau
spall_smoothing_sigma_ns <i>↑ smoother but shaves peaks/fills valleys (σ_{spall} reads low)</i>	1.5	Gaussian smoothing width (ns)
prominence_factor <i>↑ rejects weak features; ↓ catches shallow pullbacks + noise</i>	0.01	min valley/peak depth ÷ plateau
peak_distance_ns <i>↑ merges nearby features; ↓ allows closely-spaced detections</i>	3.0	min time between detected features (ns)
min_recomp_ratio <i>↑ stricter — needs a stronger recompression to accept</i>	0.015	rebound ÷ pullback depth (main gate)
min_recomp_velocity_ratio <i>↑ stricter on low-valley traces</i>	1.015	P4 ÷ P3 (rebound vs valley)
min_recomp_time_ns <i>↑ rejects fast spikes; ↓ allows quicker recompressions</i>	2.5	min rebound duration t(P4)–t(P3) (ns)

Analysis window

spall_start / end_time_ns	0 / 100	search window after t0 (ns)
threshold_velocity_ms	5.0	shock-arrival threshold for t0 (m/s)

Each parameter is illustrated on the following pages.

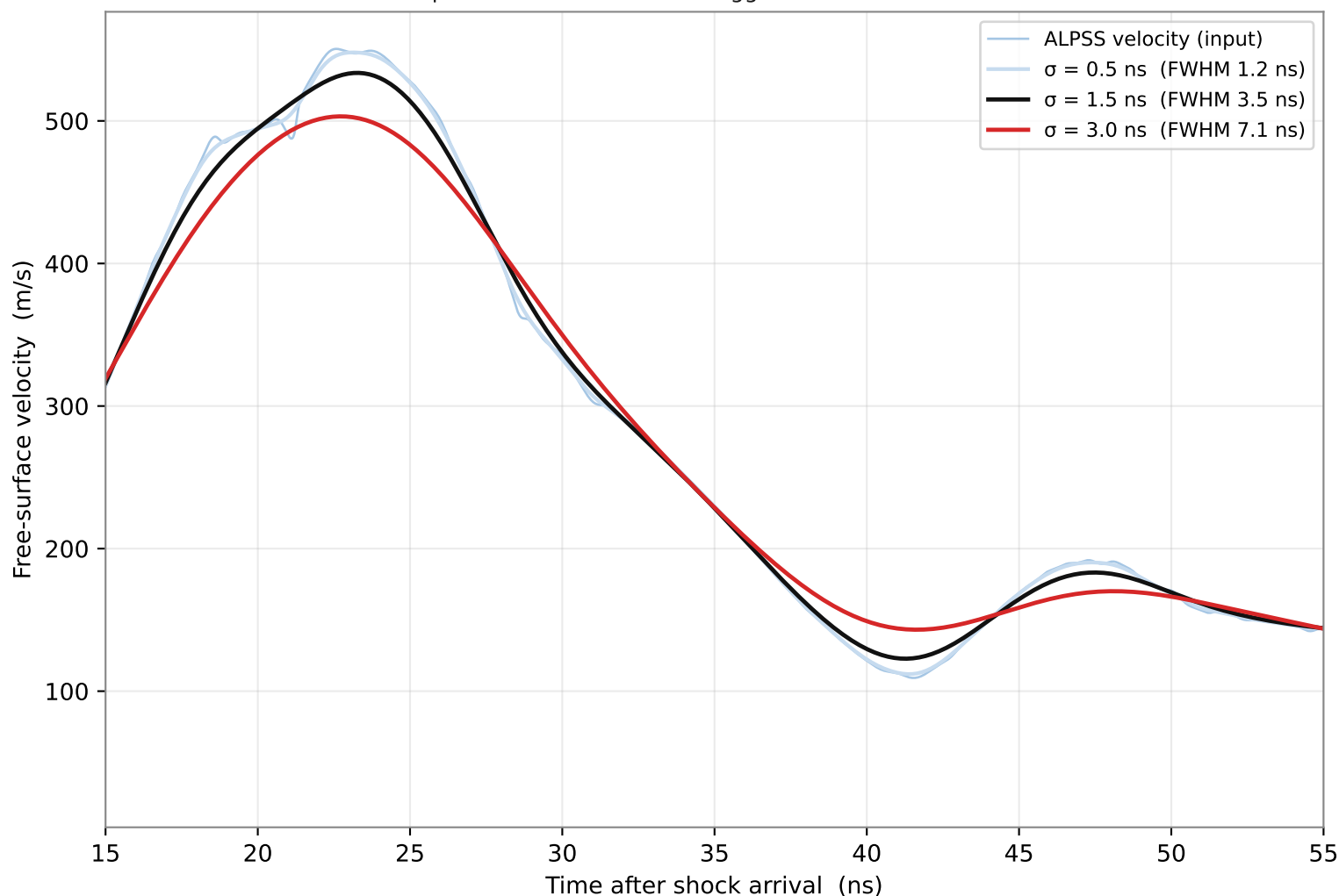
Parameter — analysis window & plateau_threshold



plateau_threshold sets the horizontal cut: every smoothed point at or above (threshold × peak) is a plateau point. Their mean is the plateau velocity that drives the shock stress and the pullback Δu (= plateau – P3). A higher value (0.99) hugs the peak — a narrow, noisier average; a lower value (0.90) widens the plateau into the rise and fall, biasing the mean downward. 0.95 is the default compromise. The window + threshold_velocity define WHERE to look and the t=0 origin; they rarely need changing unless shocks are weak or noisy.

Parameter — spall_smoothing_sigma_ns

One Gaussian pass before detection — bigger σ = smoother but flatter features



The windowed trace is Gaussian-smoothed ONCE (this parameter) before P3/P4 detection; HEL detection is unaffected. σ is in nanoseconds of real time, so the physical smoothing scale is identical on any oscilloscope sample rate.

How to choose it

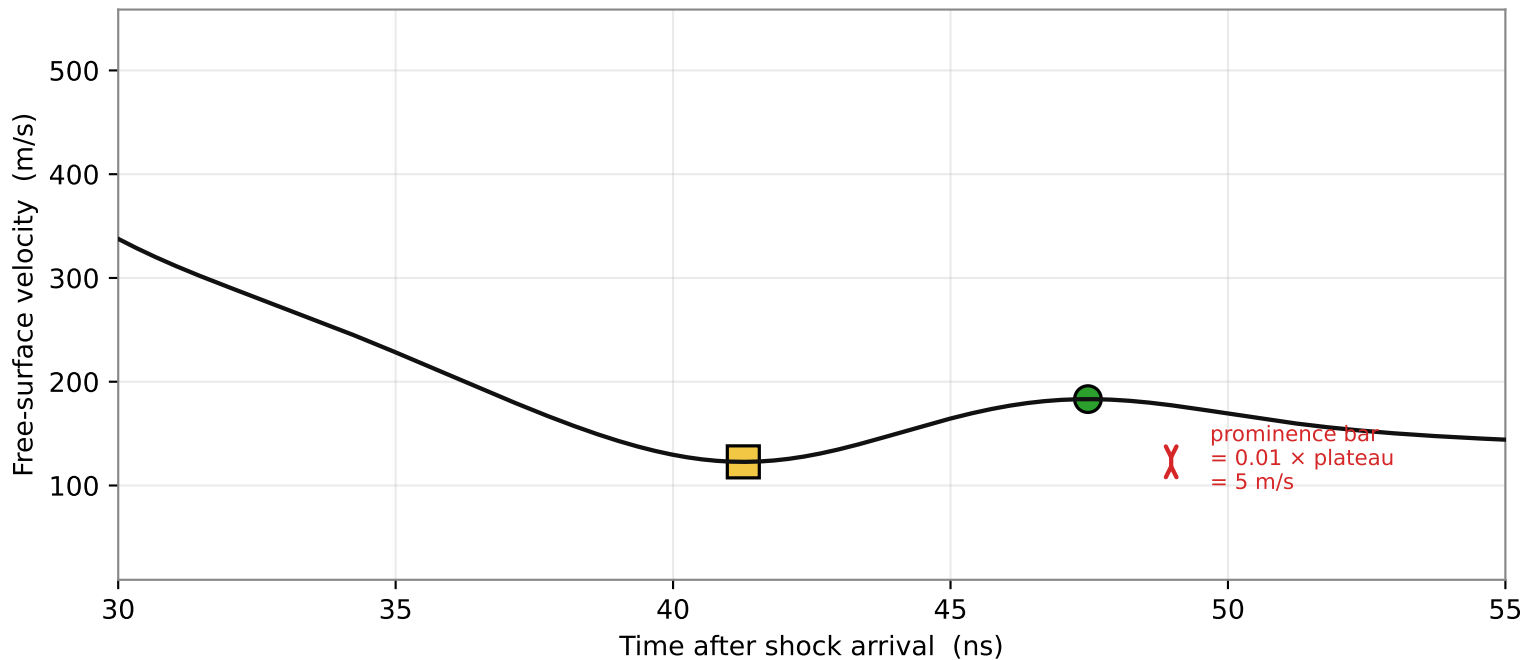
σ must sit between the noise scale (sub-ns here) and the narrowest real feature (this plateau is ~ 4 – 5 ns wide). $\text{FWHM} \approx 2.355 \sigma$:

- $\sigma = 0.5$ ns $\rightarrow$ barely smooths; noise dips can survive and mislead P3.
- $\sigma = 1.5$ ns (default) $\rightarrow$ kills sub-ns noise, keeps peak/valley amplitude.
- $\sigma = 3.0$ ns $\rightarrow$ FWHM 7 ns is wider than the plateau: it shaves the peak and fills the valley, so the pullback (and σ_{spall}) reads systematically LOW.

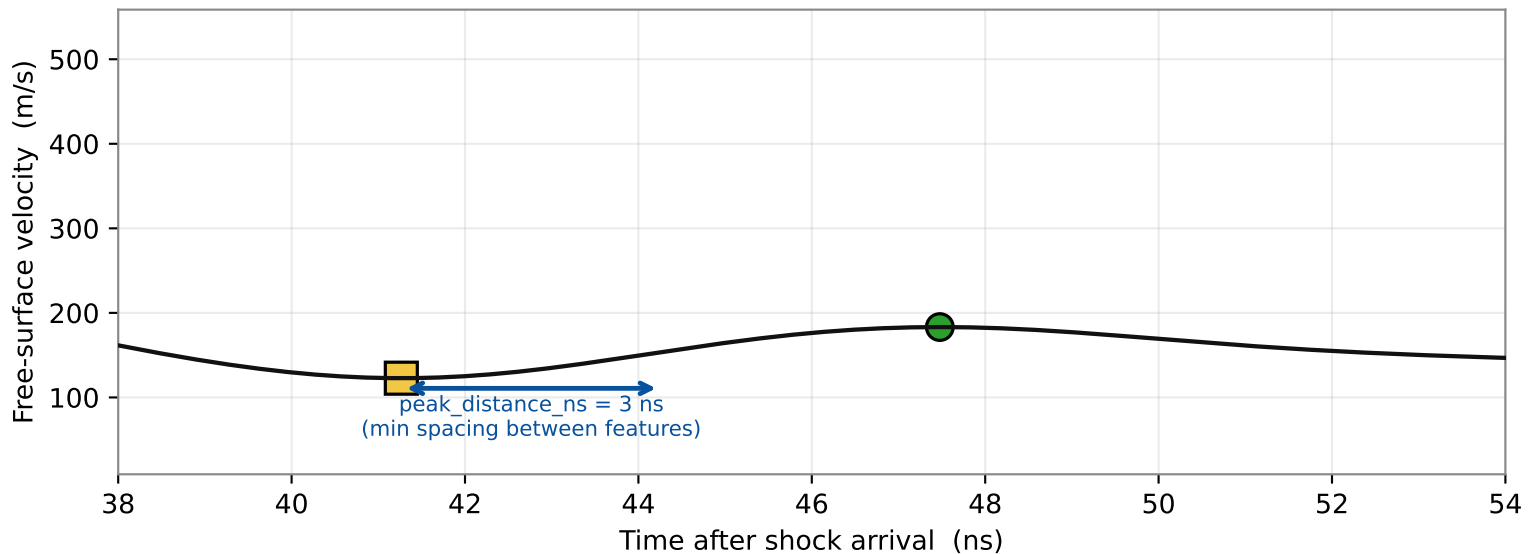
Because spall values are read from this smoothed trace, over-smoothing directly lowers the reported spall strength — keep σ as small as the noise allows.

Parameters — prominence_factor & peak_distance_ns

prominence_factor: minimum depth a dip / height a peak must have to count



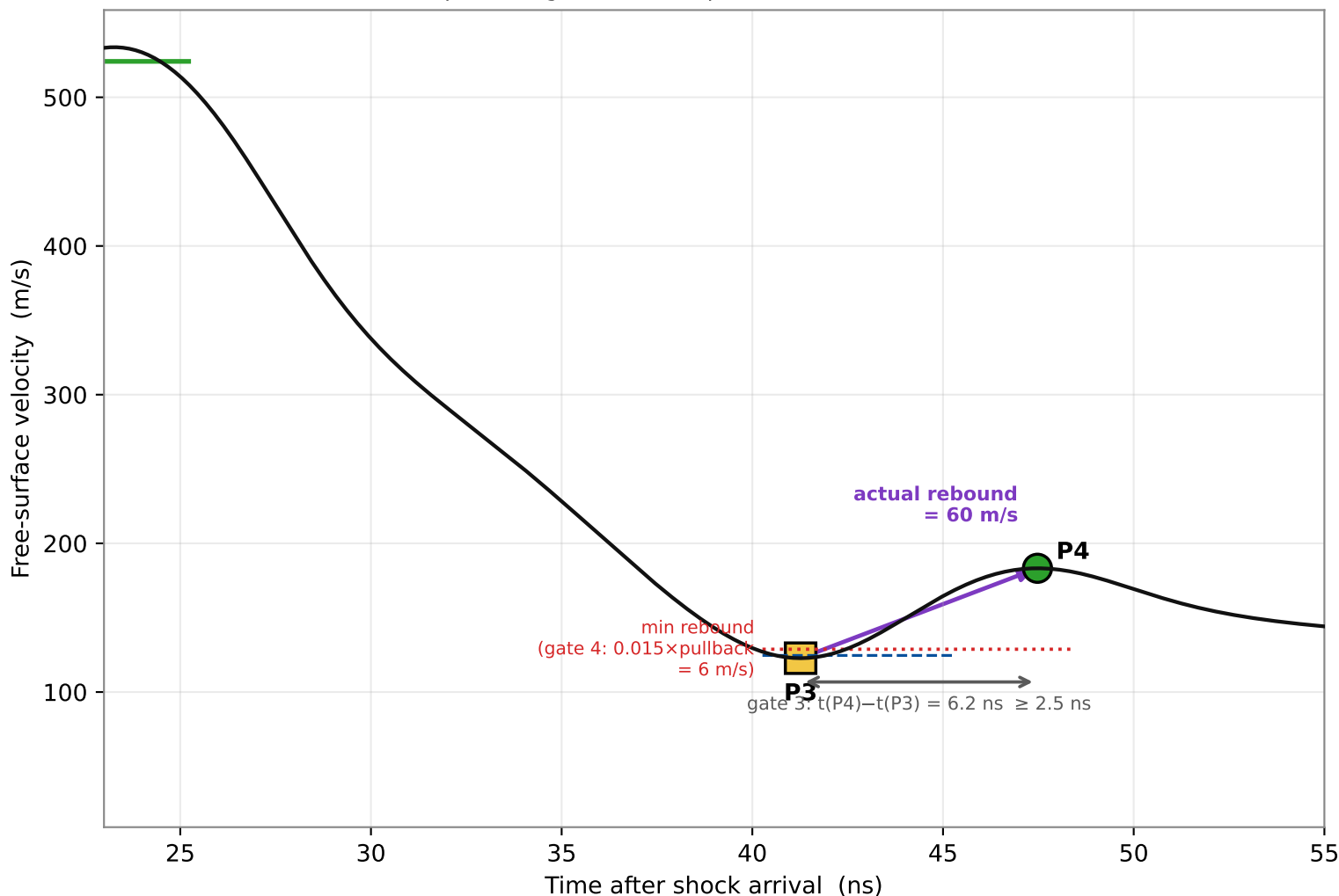
peak_distance_ns: two detected features must be at least this far apart



prominence_factor is the DETECTION floor for both the pullback valley (P3) and the recompression peak (P4): a feature must stand out from its shoulders by at least $\text{prominence_factor} \times \text{plateau}$ (here $0.01 \times 524 = 5 \text{ m/s}$). Lower it (0.005) to catch shallow/weak pullbacks; raise it (0.02) on ragged traces so a noise dimple is not detected. peak_distance_ns de-duplicates a single broad feature into one detection; keep it well below the $\sim 6 \text{ ns}$ pullback $\rightarrow$ recompression spacing so P3 and P4 are never merged.

Parameters — the three recompression gates

The recompression gates decide spall vs DNS once P3 & P4 are found



After P3 (valley) and P4 (recompression peak) are located, three thresholds confirm the rebound is a genuine spall recompression, not ringing:

min_recomp_ratio	0.015	rebound $\geq 1.5\%$ of the pullback DEPTH (main)
min_recomp_velocity_ratio	1.015	P4 $\geq 1.5\%$ above the VALLEY velocity
min_recomp_time_ns	2.5	P4 at least 2.5 ns after P3 (not a spike)

min_recomp_ratio is the primary physics gate — raise it to reject weak/ringing rebounds, lower it to keep marginal spall signals. min_recomp_velocity_ratio guards traces whose valley sits near baseline; min_recomp_time_ns rejects a fast noise spike right after the valley. Because the improved rule already requires a PROMINENT recompression peak for P3 to be accepted, these gates now act mostly as a final refinement — in practice a trace is usually resolved at the detection step (a prominent valley either has a prominent peak after it, or it does not).